Chapter 4: Cellular Respiration and Energy

Cellular respiration is the process by which biological macromolecules are broken down to produce ATP, the universal energy currency of cells.

Key Stages of Cellular Respiration:

1. Glycolysis: Occurs in cytoplasm; yields 2 ATP and 2 NADH molecules.
2. Krebs Cycle (Citric Acid Cycle): Occurs in mitochondrial matrix; yields electron carriers.
3. Electron Transport Chain: High ATP yield driven by proton gradient across inner membrane.

Chapter 5: Photosynthesis and Chloroplasts

Photosynthesis converts solar light photons into organic chemical bonds (glucose).

Light-dependent reactions take place in the thylakoid membrane.
The Calvin cycle (light-independent) fixes carbon dioxide into G3P carbohydrates.